UNIT 1: INTRODUCTION TO FISH PHYSIOLOGY

1.0 Intended Learning Outcomes

a. Identify the External and Internal anatomy of fish

b. Determine the functions of both external and internal anatomy of fish

c. Discuss the basic principles of fish anatomy and physiology in terms of locomotion, Gas exchange, Cardiovascular, osmoregulation.

1.0.1 Introduction

This learning material is a modular-based introductory overview of fish physiology which is designed to enhance student's knowledge and skills in understanding and applying the theories, concepts and principles necessary to Biology, Fish Capture, Aquaculture and other related course subjects after they will take this course.

This learning packet consists of the following parts:

- ✓ Intended Learning Outcomes (ILOs)
- ✓ Activity
- ✓ Pre-assessment
- ✓ Introduction
- ✓ Topic
  - Sub-topic
- ✓ Post-Assessment
- ✓ References
- ✓ Acknowledgement

1.0.2 Introduction to Fish Physiology

This course subject is a fundamental in taking up fisheries science, this provides students essential knowledge about the biological functions of fish and their adaptations to aquatic environments. Understanding fish physiology helps fisheries students improve fish health management, optimize aquaculture production, and develop sustainable practices.

By reading this learning packet, this will help enhancing the knowledge and skills on how to identify the parts of external/internal anatomy of fish and determining its functions which is important key to be a successful fisheries student. This will take up to 2-3 weeks on finishing this instructional material. This also includes necessary activities as well as pre-test and post-test that you need to comply.

To get the most from this module, you need to do the following:

a) Begin by reading and understanding the Intended Learning Outcomes (ILOs). These will tell you what should you know and be able to do at the end of each module.
b) Do the required Learning Activities. They begin with one or more Information Sheets. An Information Sheet contains important notes or basic information that you need to know. After reading the Information Sheet, proceed to reading of topics, this will help you ensure your mastery of basic information.
c) Demonstrate what you learned by doing laboratories answering the post-assessment which directs you to do. Your success in this learning packet material is shown in your ability to perform the required standards found in each intended learning outcome.

Each lesson also provides you with references and definitions of key terms for your guide. They can be of great help. Use them fully

In this learning packet, it covers the subsequent subtopics.

Activity # 1.

1. Illustrate and Identify the External and Internal morphology of fish

PRE-ASSESSMENT

1. What are the two (2) major classifications of fish locomotion? (5pts)
2. Differentiate stenohaline and euryhaline (5pts)
3. What are the parts of the gills of a fish? (5pts)
4. Explain how marine and freshwater fish osmoregulation works? (5pts)

1.1 GENERAL INTRODUCTION

Over half of the world's living vertebrates are fishes (more than 85 orders,536 families, 5000 living genera, 34,000 species, and counting). Fishes arose and began to radiate almost 500 million years ago and now exhibit incomparable diversity (Figure 1) in morphology, physiology, ecology, and behavior.



FIGURE 1.1 Diversity among jawless fishes, chondrichthyans, and early-branching osteichthyans.

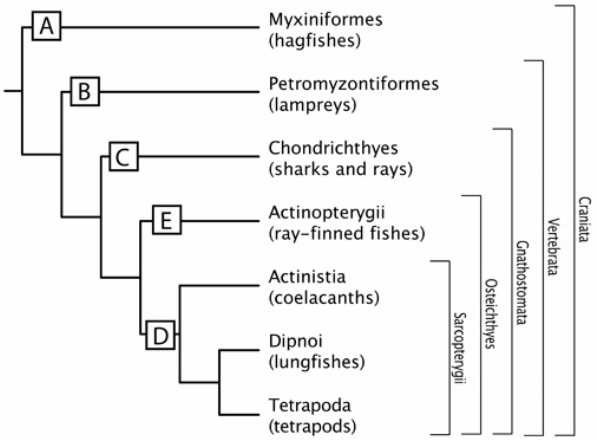


FIGURE 1.2 Phylogenetic relationships among craniates. Letters indicate the sequence of groups discussed in the text.

1.2. EXTERNAL AND INTERNAL ANATOMY OF FISH

Agnathans

Living agnathans descended from representatives of the earliest stages in the evolution of vertebrates. They lack jaws typical of other vertebrates but have a cartilaginous endoskeleton and a notochord. Seven pharyngeal gill slits are evident near the head. The gill arches separating the gill slits are reinforced with cartilage. The mouth is at the center of the round buccal funnel and is armed with horny teeth and a rasping tongue. They attach their buccal funnel to the side of a fish, rasp a hole in the body with their tongue, and feed on the body fluids of the fish.

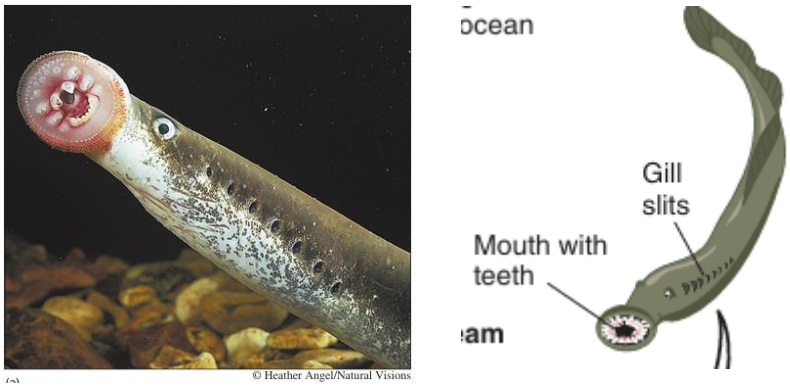


Figure 1.3. External structure and life history of a sea lamprey (class Agnatha, *Petromyzon marinus*).

Chondrichthyes

Sharks and their relatives are abundant in oceans as predators and scavengers. Their endoskeleton is cartilaginous and the anterior gill arches are modified into jaws. Jaws are a significant adaptation and modification of strong, anterior gill arches to process food (fig. 40.18). Like agnathans, their cartilaginous skeleton is not necessarily primitive but is probably derived secondarily from an ancestral bony skeleton.

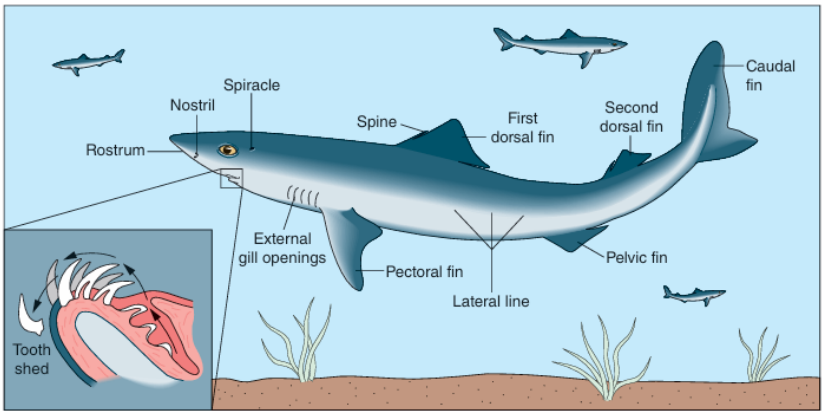


Figure 1.4. Sharks (class Chondrichthyes). Dogfish shark, *Squalus acanthias*. Section of lower jaw (inset) shows new teeth developing inside the jaw. These teeth move forward to replace lost teeth. The rate of replacement varies in different species.

Osteichthyes

Bony fish are the most diverse class of vertebrates (20,000 species). Advanced features of bony fish include a bony endo skeleton, modified gill arches, and internal air bladders for balance and buoyancy. Gills are protected by a movable gill cover called an operculum. Along each side and branching over the head of most fishes is a lateral-line system consisting of sensory pits in the skin. These pits detect water currents and predators or prey that may be moving near the fish.

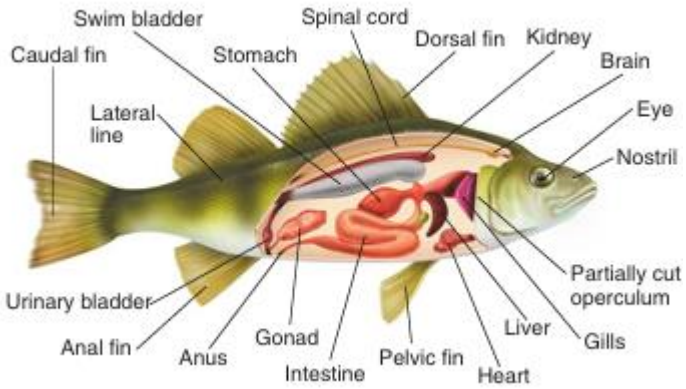


Figure 1.5. Anatomy of a bony fish, class Osteichthyes.

2. LOCOMOTION AND BIOMECHANICS

2.1 History of Fish Locomotion

Classification of swimming

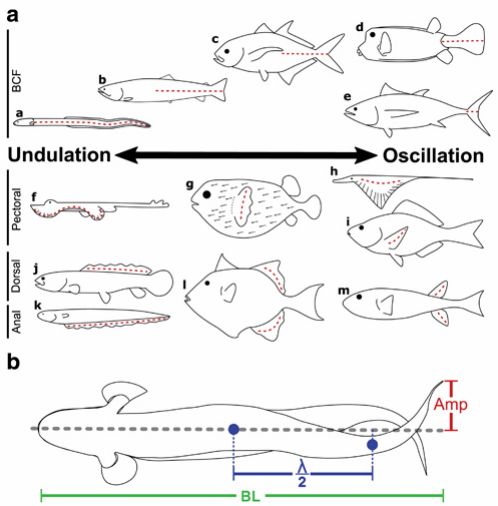


FIGURE 2.1 Diversity of fish swimming. (a) Spectrum of swimming types: a anguilliform, b – subcarangiform, c – carangiform, d – ostraciiform, e – thunniform, f rajiform, g – diodontiform, h – mobuliform, i – labriform, j – amiiform, k gymnotiform, l – balistiform, m – tetraodontiform. Red dashed line denotes the fin or body portion used in the stereotypical form of each locomotor mode. (b) Fish swimming kinematic parameters. Outline of Polypterus senegalus swimming at two points in a locomotor cycle. Tail amplitude (red), is measured from tail tip at its most lateral position to the fish's midline (approximated by dashed grey line). Body wavelength (λ) is the distance between two body wave peaks. Specific wavelength is body wavelength divided by total body length (green). (Panel A modified from Lindsey, C. C., in Fish Physiology, Academic Press, 1978, pp. 1–100.)

Fishes exhibit two broad categories of locomotion: body caudal fin (BCF) locomotion and median-paired fin (MPF) locomotion. In the former, waves pass along the fish's body, while in the latter, median (dorsal, anal) and paired (usually pectoral) fin motion provide propulsion. Fin or body motions can be oscillatory or undulatory as defined by the motion's resultant wave amplitude, wave speed and specific wavelength (body wave length/body length [BL]) (Figure 2.1).

Body Caudal Fin Locomotion

Despite large species variation, generally, body wave amplitude is smallest at the fish nose and increases posteriorly to a maximum at the tail tip. For most species, the body wave moves at a constant (or slightly accelerating) velocity towards the tail, and the specific wavelength changes between species depending on their body length, stiffness and general shape. Although the functional significance of specific wavelength is debated, it is used to help define sub-types of BCF locomotion (calculated as wavelength/body length) (Figure 2.1).

Anguilliform locomotion, used by long, thin fishes, usually has a specific wavelength of less than one (meaning that more than one wave is on the body at a time) with a relatively large lateral amplitude along most of the body (max 0.3–0.4 BL at the tail; e.g. Anguilla anguilla). **Subcarangiform** swimmers can have a specific wavelength of less than one; however, body waves occur along the posterior half or third of the body, increasing towards the tail (e.g. rainbow trout – Onchorynchus mykiss). **Carangiform** fishes usually have a specific wavelength greater than one, with waves confined to the posterior third of the fish (e.g. mackerel – Scomber scombrus). **Thunniform** swimmers (named after the Thunnus) have a specific wavelength of 1–2 BL. This relatively long propulsive wavelength is likely due to the specialized morphology of these fishes. Since they have a rigid vertebral column with myotomes (groups of muscle fibers) that insert across a larger number of vertebrae, they are less able to bend locally, thereby increasing specific wavelength. **Ostraciform** swimmers, named after the boxfishes, tend to be armored anterior to the caudal fin, and so often only the caudal fin undulates (Lindsey 1978). These fish can use synchronous

movements of their pectoral, dorsal, anal and caudal fins depending on their speed. For example, pectoral and anal fins power swimming at low speeds (<1 BL/s), anal and dorsal fins power swimming between 1 and 5 BL/s, and a caudal, kick-and-glide gait is used above 5 BL/s.

Many of the variables used to describe and quantify swimming are in two dimensions. Three-dimensional data on fishes is complex, and although relatively easy to acquire with modern high-speed video, remains difficult to quantify meaningfully. Recent work shows that the three-dimensional (3D) motion or twist of fishes may also be important, particularly in elongate fishes. Computational fluid dynamists who integrate the 3-D shape and motion of fins and body to describe predicted flow patterns produced by swimming fishes are showing that a 3-D approach to fish swimming is essential to determine how unsteady movements influence fish locomotion. While the body and/or caudal fin are likely responsible for producing the majority of the thrust during BCF swimming, the fins are under active muscle control to help stabilize against perturbations that might otherwise cause roll, tilt or yaw.

Gait Changes

A fish may exhibit one type of swimming at slow speeds and transition into another at higher speeds. For example, the bluegill sunfish oscillates its pectoral fins at speeds under 1.0 BL/s and changes to BCF swimming at higher speeds. In fact, most fishes (especially BCF swimmers) will use the "kick and-glide" mode of locomotion when swimming at very high speeds. This behavior is composed of short bursts of caudal fin oscillation interspersed with glides where the body is straight, presumably to reduce drag. At high swimming speeds, this gait appears to be more energy efficient than other modes of steady swimming. Different swimming modes may also be more or less efficient depending on the environment. For example, in an experiment where fishes were exposed to water flows that mimic coral reef wave surges, it appeared that MPF swimmers were better able to reduce costs associated with increased wave surge frequency compared with BCF swimmers.

Median and Paired Fin Locomotion

There is a diverse spectrum of median and paired fin (MPF) use, from undulatory motion (passing a wave along the fin surface) to oscillatory motion (rowing the fin back and forth). Fin motions often change depending on swimming speed, and individuals can switch between pectoral fin, median fin, and body and caudal fin use as required. MPF swimming modes are generally classified by the group of animals that tend to demonstrate the behavior. **Labriform** locomotion is powered by oscillatory pectoral fin flapping (Figure 2.2) and can be drag-based or lift-based. The drag-based form consists of a power stroke followed by a recovery stroke and is used across a narrow range of swim speeds by a variety of fishes, from short-bodied Centrarchidae and Cichlidae to long-bodied Nototheniidae. During the power stroke, fins adduct through the water with their maximum surface area perpendicular to the axis of propulsion. During

recovery, fins rotate and abduct with the maximum surface area parallel to the direction of travel to minimize drag. In contrast, lift-based labriform swimming orients the fin parallel to the flow and oscillates it in a dorso-ventral motion to create lift (Figure 2.2. Labridae and perciform fishes (i.e. Embiotocidae; use this form of motion; however, drag and lift-based modes are often used in conjunction with each other depending on the needs of the animal. The oddly shaped pectoral fins of skates and rays can also oscillate in what looks like underwater flapping flight (**Mobuliform** locomotion. Dorsal and anal fins can also be used in an oscillatory fashion (**Tetraodontiform** locomotion), as seen in the pufferfishes that flap their dorsal and anal fins from side to side in concert to power locomotion.

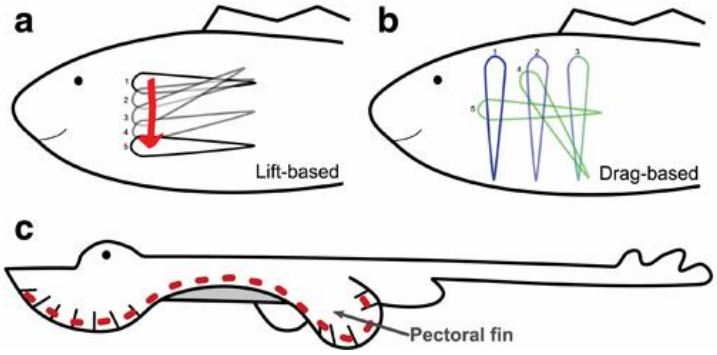


FIGURE 2.2 Pectoral fin swimming. Labriform swimming. (a) Lift-based, dorso ventral fin motion acts like a flapping foil to produce lift and thrust. (b) Drag-based: fin adduction during the power stroke creates thrust (steps 1–3); during the recovery stroke, the fin is abducted parallel to the flow, minimizing drag (steps 3–5). (c) Rajiform pectoral fin undulation passes a wave anterior to posterior along the fin edge. (Panels a and b modified from Walker and Westneat 2002a.)

Undulatory fin motion involves passing waves along the surface of the median fins (**Gymnotiform** – anal fin, **Amiiform** – dorsal fin, and **Balistiform** – both anal and dorsal fins) or the pectoral fins (**Rajiform** and **Diodontiform** locomotion) (Figure 2.1). Generally, during these forms of steady swimming, the body is straight, and additional fins are not used or are entirely absent. One notable characteristic of this form of swimming is the ability to swim both forward and backward with fluidity. Many species combine oscillatory and undulatory fin movements, the locomotion style correlating with the ecology and morphology of the fish.

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2.2 COMPLEXITY OF FISH FORCES

General Biomechanics: Force, Power and Thrust

Before understanding how fishes produce forces and the consequences of those forces, it is important to clarify the basic relationship between force, work and power. Force is the product of an object's mass and its acceleration (F = ma with units kgm/s2 or N); therefore, the size and acceleration of an animal will determine the amount of force it must produce. The amount of work an animal performs, which may be considered a more important evolutionary metric, is the force produced times the distance travelled; W = F*d with units kgm2/s2 or J. Finally, power is the amount of work done per unit time; P = W/t with unit's J/s or W. From an organismal perspective, you can consider that an animal experiences mechanical forces from the environment and produces biological forces with its muscles. Muscles can produce positive, negative or no work depending on the resultant motion of the muscles and/or animal, and finally, the speed with which the animal can perform the work will determine the power it can produce. Using a whole animal as an example, a fish accelerating from zero velocity in still water produces a force with its tail (mass × acceleration). The distance it travels with that amount of force determines the work it produces (F × d). The time it takes for it to cover that distance or complete the work tells you the power it took (W/t). This type of motion analysis can also be done at the muscle level, thinking about how far and how fast myosin and actin move relative to each other while producing force. With force, work and power concepts firmly in our minds, we can now discuss how fishes produce forces and what it means in evolutionary terms from the tissue level to the whole organism.

Muscle Anatomy

How the muscle fibers attach to the skeleton is also exceptional in fishes and leads to an interesting passing of forces along the body. Fish body musculature is divided by muscle fiber type; the red muscle is located in a very small band along the side of the fish, and it is used for routine, continuous swimming Figures 2.4 and 2.5). The bulk of the body is white muscle, used for powerful maneuvers, burst swimming or escape response, and consists of a series of nested units. Each unit is called a myomere, and they attach intricately with one another in a V or W nested cone configuration (Figure 2.4). A series of inter myomere tendons and fascial sheets allow the contracting muscle to pass forces between the spinal vertebrae and the fish skin, as well as along the cones of the myomeres themselves, resulting in body stiffening and passing forces to augment tail motion. The unique structure of the myomeres in fishes, as well as the

curved trajectories of the individual muscle fibers within the myomeres, has been explained via the uniform strain hypothesis, which suggests that this orientation allows all muscle fibers of a given type to contribute equally to force production during swimming. Because a swimming fish is essentially a bending beam, and because beam theory states that forces are greater the further you are from the neutral axis of the beam, if fish muscle ran parallel with the spine, the fibers located furthest from the spine would provide the most force, and those close to the spine would provide none. The uniform strain hypothesis states that for a fish to effectively use all of its muscle fibers, they must alter their orientation with respect to the spine. Subsequent modelling supports this idea by accurately predicting the trajectories seen in fish muscle fibers based on achieving mechanical stability of the fish.

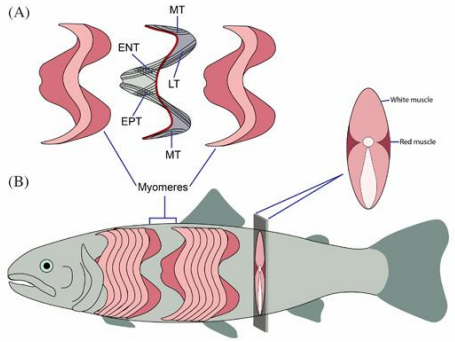


FIGURE 2.4 Fish muscle anatomy. (A) The arrangement of myomeres in a gnathostome. (B) Diagram showing how myomeres (pink) and myosepta (grey) nestle into one another. Tendons as part of the myosepta (grey) are indicated: epineural tendon (ENT), epipleural tendon (EPT), myorhabdoid tendons (MT) and lateral bands/tendons (LT). The attachment line of the myosepta to the skin is indicated by the red line. (Tendon alignment adapted from Gemballa, S., et al., Journal of Evolutionary Biology, 16, 966–975, 2003; trout myomere image adapted from Hickman, C. L., et al., 2018. Laboratory Studies in Animal Diversity, 8th ed, McGraw-Hill Publishing, 2018.)

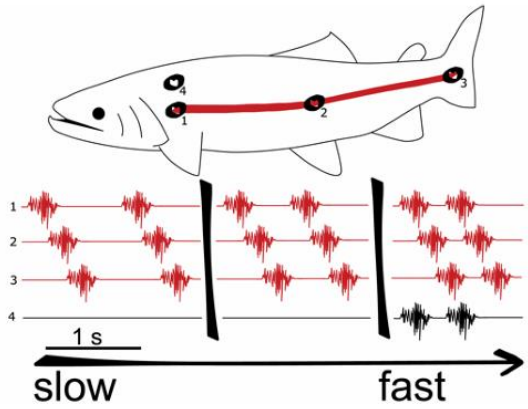


FIGURE 2.5 Muscle recruitment during swimming. Red muscle is active along the length of a fish at all swim speeds and increases contraction frequency with speed. White muscle is recruited at higher swimming speeds. (Modified from Moyes, C.D. and Schulte, P.M., Principles of Animal Physiology, 2nd ed., Pearson, 2016.)

The anatomy of the fish musculature, in combination with the tuning capabilities of the muscle fibers themselves (white, red and intermediate fibers allows fishes to engage all of their muscle mass for powerful burst swimming or only part of their muscle mass for moderate endurance swimming. White and red fibers are more heterogeneously mixed in fin muscles; however, there is often a general pattern of centrally located aerobic fibers with surrounding white fibers. The functional implication of fiber regionalization in fins is not clear.

Diversity of fin anatomy and structure

Most fishes have two sets of paired fins, the pectoral and the pelvic fins, and three median fins, the dorsal, anal and caudal fin (Figure 2.1). During the evolution of fishes, immense variation in fin number, shape and size has occurred. Many of the functions of ornate and elongate fins are for purposes other than locomotion, such as sexual selection (i.e. swordtails) or crypsis (i.e. sea dragons), and there is often a tradeoff between alternate functions and locomotion. Among the fins selected for locomotory performance, there are many shapes, sizes and levels of stiffness, and each is used in a different style of locomotion. Although there are differences in the degree of fin element stiffness or fusion, they are all controlled in a similar way. Fins are composed of segmented boney elements, or fin rays, that are joined together by a flexible membrane (Figure 2.6). Each fin ray, termed a lepidotrichia, is made up of two hemitrichia. These bones are joined by elastic elements, which allows them to slide relative to one another and facilitates bending of the fin. A set of muscles at the base of each fin ray controls the motion of the ray, moving it relative to the body but also moving hemitrichia relative to each other, allowing the fin to bend. In general, fish fins are incredibly deformable surfaces that are highly controlled. Fishes can increase or decrease fin surface area depending on the required force production and function. Fins are critical for stabilization and destabilization during locomotion and maneuvering.

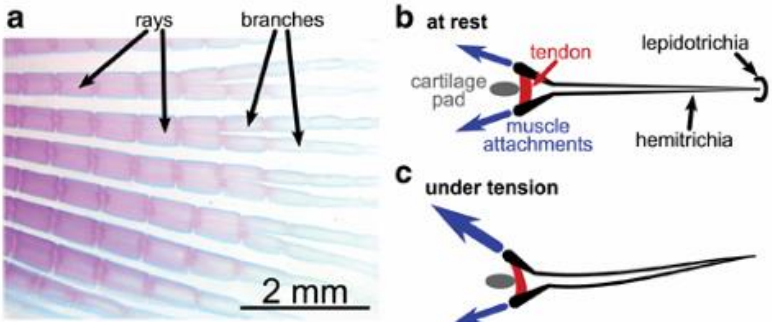


FIGURE 2.6 Fish fin rays are under active muscle control to change fin shape. (a) Fin rays are segmented and branch near the tip, as can be seen in a cleared and stained fin. At rest (b), the two hemitrichia align, and the lepidotrichia is straight. (c) Under unequal muscle tension, the hemitrichia slide relative to each other, thereby bending the ray. (Modified from Alben, S., et al. Journal of The Royal Society Interface, 4, 243–256, 2007.)

2.3 MUSCLE ACTIVITY AND NEUROCONTROL

Muscle activity

Swimming speed increases with tailbeat frequency, driven by an increase in muscle recruitment and an increase in the frequency of the muscle activation wave that passes along the body (Figure 2.5). The timing of muscle strain cycles (the lengthening and shortening of muscle fibers) relative to muscle activity is essential to the functional performance of a muscle during. The state of the muscle fiber (its length and motion) before activation will influence the amount of force it can produce (Figure 2.3). Thus, the difference in timing of muscle activation within a strain cycle (i.e. the phase lag) is critical and varies both with species and along the length of a. At low speed, locomotion is powered exclusively by red muscle, and white muscle is progressively recruited as fish swim faster (Figure 2.5).

Bcf swimming

How red muscle is used during different swimming modes is still debated. The larger-amplitude wave that is found all along the body in anguilliform swimmers appears to be accompanied by more uniform head to tail muscle activity patterns. In contrast, the anterior body muscle contractions of non-anguilliform swimmers do not accompany an anterior body wave; they produce forces that pass via stiffened muscle and tendon to the posterior body wave and tail to be subsequently turned into thrust. The suggested functional outcome of this, which is supported by the hydrodynamic forces produced in the wake of fishes, is that anguilliform swimmers produce power and thrust along the entire length of their bodies. The passing of muscle effort posteriorly in non-anguilliform swimmers results in differences in muscle timing and behavior along the fish length that are much more noticeable. For example, as you approach the tail of the fish, there is a decrease in the duty cycle, the amount of time the muscle is on during a tailbeat (anguilliform swimmers have a constant duty cycle along their length). The amount by which a muscle shortens (strain) and the resultant body amplitude increase towards the tail. The phase lag in timing between muscle activation and resultant body amplitude also increases towards the tail (posterior muscles come on earlier relative to peak wave amplitude) (Figure 2.7). In essence, the wave of muscle activity passes anterior to posterior faster than the wave of body deflection. This is taken to an extreme in tunniform swimmers, where the wave of muscle activity passes along the anterior of a tuna so quickly that it is near synchronous, straightening and stiffening the anterior body and passing all of its force to caudal thrust production. Changes in muscle function along the fish body can be related to swimming speed as well as the ecology of the fish.

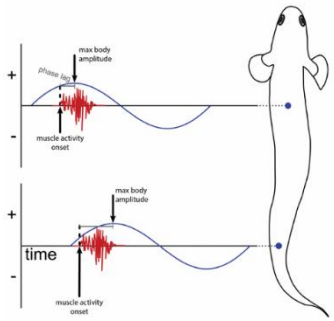


FIGURE 2.7 Phase lag between muscle activation and body kinematic event. Maximum body amplitude occurs sooner following muscle activation at the fish's anterior than at its posterior. Muscle position is denoted as a blue dot on the fish outline.

Labriform locomotion

Much less work has been done on the muscle activation patterns in fin musculature during swimming. This is partly due to the very small size of many fin muscles and the difficulty in accurately implanting electrodes without damaging muscle function. Most work is on the anatomical function or free-swimming activation patterns in the pectoral fins of labriform swimmers. Although the large abductor and adductor muscles tend to be responsible for the larger-scale fin motions, a variable number of smaller fin muscles also contribute to the locomotor stroke. The gait transition from pectoral fin swimming to BCF swimming is likely due to aerobic fin muscles reaching a limit in their power output, necessitating the recruitment of body muscle to swim faster.

Unsteady swimming

All the above has been concerned with fishes swimming steadily, but of course, most fishes do not swim steadily, and most of their behavior in the wild is composed of turns, escapes from predators, swimming through turbulent flow, etc. Limiting studies to steady swimming simplifies the system and allows basic observations of muscle patterns, kinematics and neural control; however, understanding the full range of kinematics and muscle activity that fishes use on a regular basis requires thorough quantification of unsteady swimming.

Escape response

One of the best-studied unsteady behaviors that fishes perform is the fast start. Fast starts can be initiated either by the reticulospinal neurons (a slower response) or by the Mauthner neurons (a faster response) to escape predators and catch prey. This behavior is composed of three kinematic stages: preparatory stage 1, propulsive stage 2 and variable stage 3. Stage 1 is generally defined as the formation of the initial starting position (either a "c" shape or an "S" shape). Stage 2 can be defined in a variety of ways but generally corresponds to the change in direction of the anterior body, the onset of forward propulsion, and the onset of contralateral muscle activity (though each of these occur at slightly different times). Stage 3 is highly variable and loosely defined. Unlike steady swimming, the neural control that leads to a Mauthner fast start is relatively well established. Stage 1 is initiated by the triggering of the Mauthner neurons, which are connected to each other via gap junctions, direct cell–cell electrical connections that allow rapid transmission of neural signals. This neuroanatomy facilitates extremely rapid initiation of escape responses via sensory feedback.

Swimming in unsteady flow

While it might seem as though swimming through water with increased or turbulent flow would pose an inherent problem to fishes, in fact, they can gain some

energetic advantage by swimming strategically through vortices in water. When objects obstruct flowing water, the water bends around the object and creates vortical flow that sheds in a characteristic pattern (the von Karman Street). A series of experiments that swam fishes in flow behind a D-cylinder have demonstrated that, rather than fighting against the structure's wake, fishes can actually use the regular vortical flow to reduce their energy expenditure, decreasing muscle activity while holding station. In fact, slaloming between the vortices that form behind a cylinder is so effective at capturing energy from the flow that even a dead trout, pointing into the flow, can passively "swim" upstream in the wake of a structure.

Neuro control

Motor neurons send signals to the muscles to coordinate muscle contraction, which leads to locomotion. The pattern of the motor signals can be the result of a "top-down" signal from the brain and/or rhythmical signals from local central pattern generators (CPGs), both of which can be influenced by sensory feedback experienced by the fish as it swims through its environment. Our understanding of the importance and control of CPGs has improved since the initial descriptions that locomotion may be controlled by the spinal cord. It is now known that CPGs are responsible for controlling a variety of behaviours, including breathing, chewing, swallowing and locomotion. Locomotor CPGs are neural centers found in the spinal cord that can provide rhythmic motor output to the muscles, which is tuned via sensory and top-down input from a heterogeneous environment.

Much of our knowledge of CPG function is from work done on the lamprey. The swimming CPGs of the lamprey are along the entire length of the spinal cord, and an isolated spinal cord can produce motor patterns that closely resemble that of an intact lamprey. This motor pattern is only created in response to an excitatory drive (e.g. brainstem stimulation, sensory stimulation or excitatory amino acids). The presence of CPGs has also been demonstrated in spiny dogfish, stingray, carp. While these CPGs create the intrinsic rhythm of swimming, they need additional input in order to adjust the fish's behaviour appropriately. Usually, the excitatory drive to turn on the CPGs is provided by the mesencephalic locomotor region (MLR) and passed via the reticulospinal neurons to the spinal CPGs. The MLR also receives top-down inputs, such as vision or hearing, from the cerebral cortex (pallium in fishes). While these inputs historically are poorly characterized, recent work has shed some light on the pathway that carries olfactory information to the MLR. This sensory input appears to be able to influence the MLR output (and thus the resulting locomotor behavior) in a graded way, thereby fine-tuning an animal's behavior. Proprioception in fishes has only recently started to become known, showing clear sensory feedback associated with fin use in fishes.

2.4 AMPHIBIOUS LOCOMOTION IN FISHES

Diversity of terrestrial locomotion

There are over 200 different species of extant amphibious fishes representing at least 17 genera. The diversity and range of morphological and behavioral phenotypes found among these fishes are large. There are many challenges associated with moving from an aquatic to a terrestrial environment: desiccation, gas exchange, pH balance and disposal of nitrogenous wastes are just a few. Each of these affect's locomotion indirectly by influencing muscle and tissue performance. Here, we will focus on the impacts a transition from an aquatic to a terrestrial environment has on the locomotor performance of an animal.

The mechanical environments of water and land are very different. Viscosity and drag are the major forces experienced by fishes in water, whereas gravity and friction become the major challenges on land. Fish body and fin function must be flexible in order to produce appropriate forces across environments. Different fishes have different strategies when moving about on land. There are three main categories of fish terrestrial locomotion axial-based locomotion, where motion is produced entirely by the body (i.e. snake-like locomotion in elongate ropefish or eel, or jumping locomotion in blennies and rivulus); appendicular-based locomotion, where motion is derived entirely from the fin (i.e. mudskipper); and axial-appendicular-based locomotion, where the animal uses both fins and body to produce thrust (i.e. walking catfish and Polypterus). Some amphibious fishes have totally different strategies, such as climbing perch that use modified opercula to walk or the waterfall climbing gobies that use their mouths as suckers to cling to rockfaces under rushing flow.

Often, fishes exhibit very different behaviors in one environment relative to the other. For example, fin range of motion increases during walking in both bichir (*Polypterus senegalus*) and mudskipper (*Periophthalmus argentilineatus*); however, body oscillation increases for bichir and decreases for mudskipper (Figure 2.8). These differences between species illustrate that there are multiple ways to modify strategies for locomotion across environments.

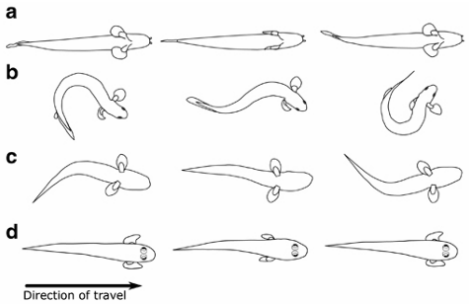


FIGURE 2.8 Aquatic vs. terrestrial locomotion. (a) Polypterus senegalus swimming in water keeps the body relatively straight and uses fins to provide thrust; ventral view. (b) P. senegalus walking on land uses exaggerated body movements and alternating fin plants; dorsal view. (c) Periophthalmus argentilineatus swimming in water incorporates body bending with reduced fin motion; ventral view. (d) P. argentilineatus walking on land keeps body straight, and fins are planted simultaneously; dorsal view. (Panels a and b modified from Standen, E. M., et al., Nature, 513 54–58, 2014; c and d

modified from Pace, C. M. and Gibb, A. C., Journal of Experimental Biology, 212, 2279–2286, 2009.)



FIGURE 2.9 Amphibious fishes. Left, the walking Catfish (*Clarias batrachus*) which is locally known as **Pantat** in Samar, Philippines. Right, Barred Mudskippers (*Periophthalmus argentilineatus)* are known climbing out of water at mangrove and mudflats area. Photos by: U.S. Geological Survey and Randall, J.E. respectively.

3. GAS EXCHANGE

Oxygen and carbon dioxide dissolve in water, and most fishes exchange dissolved oxygen and carbon dioxide in water by means of the gills. The gills lie behind and to the side of the mouth cavity and consist of fleshy filaments supported by the gill arches and filled with blood vessels, which give gills a bright red color. Water taken in continuously through the mouth passes backward between the gill bars and over the gill filaments, where the exchange of gases takes place. The gills are protected by a gill cover in teleost and many other fishes but by flaps of skin in sharks, rays, and some of the older fossil fish groups. The blood capillaries in the gill filaments are close to the gill surface to take up oxygen from the water and to give up excess carbon dioxide to the water.

Most modern fishes have a hydrostatic (ballast) organ, called the swim bladder, that lies in the body cavity just below the kidney and above the stomach and intestine. It originated as a diverticulum of the digestive canal. In advanced teleost, especially the acanthopterygians, the bladder has lost its connection with the digestive tract, a condition called physoclistic. The connection has been retained (physostomous) by many relatively primitive teleosts. In several unrelated lines of fishes, the bladder has become specialized as a lung or, at least, as a highly vascularized accessory breathing organ. Some fishes with such accessory organs are obligate air breathers and will drown if denied access to the surface, even in well-oxygenated water. Fishes with a hydrostatic form of swim bladder can control their depth by regulating the amount of gas in the bladder. The gas, mostly oxygen, is secreted into the bladder by special glands, rendering the fish more buoyant; the gas is absorbed into the bloodstream by another

special organ, reducing the overall buoyancy and allowing the fish to sink. Some deep-sea fishes may have oils, rather than gas, in the bladder. Other deep-sea and some bottom-living forms have much-reduced swim bladders or have lost the organ entirely.

The swim bladder of fishes follows the same developmental pattern as the lungs of land vertebrates. There is no doubt that the two structures have the same historical origin in primitive fishes. More or less intermediate forms still survive among the more primitive types of fishes, such as the lungfishes Lepidosiren and Protopterus.

Ventilation

Most species, as adults, possess four to seven gill arches on each side of the head that are collectively referred to as the branchial basket. The developmental timeline for gill formation and, in some species, the external to internal migration of the structures depend on myriad factors; as well as species and temperature, other abiotic factors may be influential. Water must pass over the gills via ventilatory flow rate (V. w), which is largely the same process among species. For most teleost fishes, ventilation involves asynchronous buccal (mouth) and opercular (gill cover) pumping to generate pressure. In contrast, ram-ventilators, such as some pelagic teleosts and many elasmobranch species, enerate dynamic pressure while swimming, which drives water over the gills. Some species may transition from buccal pumping to ram ventilating, with a change in swimming speed and/or environmental O2 level likely due to the energetic savings associated with gill water flow being powered by skeletal muscles rather than by the buccal pump muscles.

Morphology

From a physical perspective, gill filaments and lamellae can act like a sieve, being numerous or even fused at the tips to maximize gas exchange with the water. Indeed, the morphology of the gill filaments and lamellae and how they contact the external environment have been well studied across species, lifestyles, habitat, and developmental stages, as histological and microscopy preparations have been long established. On each filament, lamellae are comprised of two epithelia separated by pillar cells for structural integrity and to enable blood flow through the lamellae (Figure 3.1).

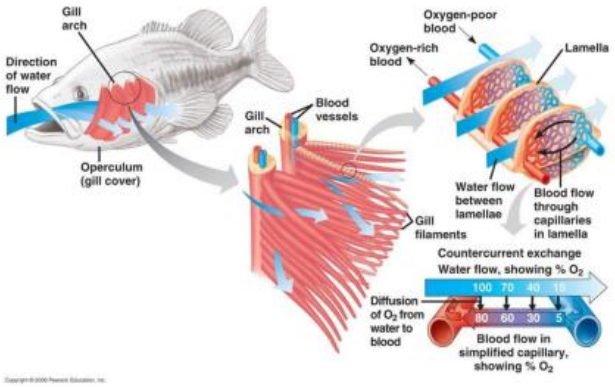


Figure 3.1. Schematic of multi scale architecture of fish gill demonstrating exchange of oxygen from ambient water to fish body

4. The Cardiovascular System

General introduction

The cardiorespiratory system of fishes, as for other vertebrates, is designed to efficiently transport a plethora of substances and molecules, including respiratory gases, hormones, metabolites, electrolytes, nutrients, plasma proteins, immune factors, signaling molecules, and wastes, among the tissues. The fish cardiovascular system is composed of four fundamental components: blood; heart(s); vasculature; and a control system. This chapter broadly describes the general features of the fish cardiovascular system, highlighting the latest ideas, advances, and research in the field, and details some of the integrated responses of this system to changes in ambient and internal conditions.

General features of the fish cardiovascular system

Blood

Blood serves to transport substances throughout the body. It is composed of red blood cells (RBCs, erythrocytes), white blood cells (WBCs, leukocytes), and plasma (extracellular fluid that contains proteins, ions, hormones, nutrients, metabolites, and wastes). The total blood volume is in the range of 3–4% of body mass in most teleost fishes and 5–8% of body mass in most elasmobranchs. Antarctic icefishes (Nototheninoid fishes from the family Channichthyidae) do not have RBCs, and their blood volume is approximately four times higher than that of typical fishes. Hagfishes top all known fishes, with a blood volume of ~18% of body mass. RBCs serve to transport oxygen and carbon dioxide and regulate acid base balance. In contrast to mammals, fish RBCs are nucleated and thus, can synthesize proteins in the circulation. Fish blood is red (except in icefishes) because their RBCs contain hemoglobin. The hemoglobins of fishes are tetrameric (built of four polypeptide chains), with the exception of hagfishes and lampreys, which have monomeric and dimeric hemoglobins. Hematocrit (% of blood volume that is occupied by RBCs) varies widely across fishes, from 0% to >50%, though in most teleosts, it tends to range from 20% to 40%. Blood viscosity increases with hematocrit and cold temperatures, which results in high vascular resistance and high cardiac workload. Accordingly, fishes in cold, stable environments often have a lower hematocrit, which serves to decrease blood viscosity The primary role of WBCs is in immunity and cellular defense. Leucocrit (% of blood volume that is occupied by WBCs) constitutes a tiny proportion of blood volume (0.3–1.0%). There are four major types of WBCs: lymphocytes, thrombocytes, granulocytes, and monocytes. Lymphocytes are common and are involved in immuno-competence and antibody production. Thrombocytes are small and play a role in blood clotting. The function of granulocytes is unknown, but they are thought to play a diversity of roles,

including in the inflammatory response, blood coagulation, tissue repair, and phagocytizing microorganisms. The largest and rarest WBCs are the monocytes, which are phagocytic (ingesting harmful cells, particles, and bacteria). Infection and parasitic infestation typically result in an increase in circulating WBC levels, while stress and environmental toxicants can modulate levels and thus influence disease resistance. The components of the blood are tightly regulated; thus, blood sampling is a common technique to assess the physiological status of a fish. Numerous hormones are transported in the plasma (e.g., gonadotropins and sex steroids, growth hormone, thyroid hormones, and cortisol). Metabolite levels can be highly variable in the plasma. Lactate, the end product of anaerobic metabolism, accumulates in the blood following strenuous activity (e.g., exhaustive exercise) and in response to oxygen limitation (hypoxia: decreased oxygen; anoxia: no oxygen). Anoxia-tolerant fishes such as crucian carp (Carassius carassius), and its close relative the goldfish (Carassius auratus), produce ethanol in response to severe hypoxia or anoxia. The major plasma electrolytes in fishes are Na+ and Cl−, whereas plasma Mg2+, K+, and Ca2+ are found at much lower levels. Amino acids are transported within the RBCs and in the plasma, while lipoproteins transport lipids in the plasma. Several plasma proteins are present in the blood (including albumin, Ca2+-binding proteins, and immunoglobulins). In addition, some teleosts in polar regions can produce antifreeze proteins or antifreeze glycoproteins to inhibit ice crystallization within body fluids that would otherwise be fatal.

Heart morphology and blood flow patterns

The fish heart is composed of morphologically distinct regions (segments). Classically, the fish heart was considered to be composed of four chambers arranged in series: the sinus venosus, atrium, ventricle, and bulbus (in teleosts; or conus in elasmobranchs) arteriosus. However, the current consensus is that the fish heart consists of six components (or segments) arranged in series within the pericardium. These are the sinus venosus, atrium, atrioventricular (AV) segment, ventricle, conus arteriosus, and bulbus arteriosus, the latter two of which are collectively termed the outflow tract (OFT) (Icardo 2017) (Figure 4.1). The Cyclostomata (hagfish and lampreys) present an exception. In these species, all components of the OFT are absent.

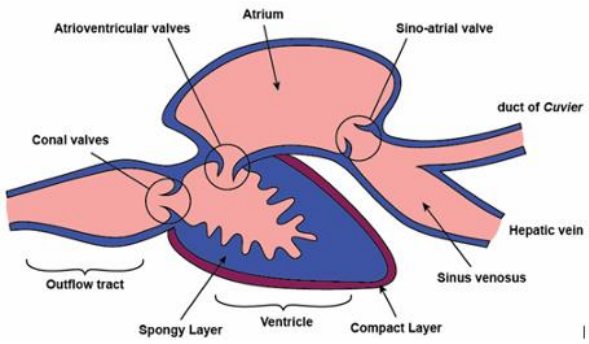


FIGURE 4.1 Anatomical organization of the teleost heart. Direction of blood flow is from right to left.

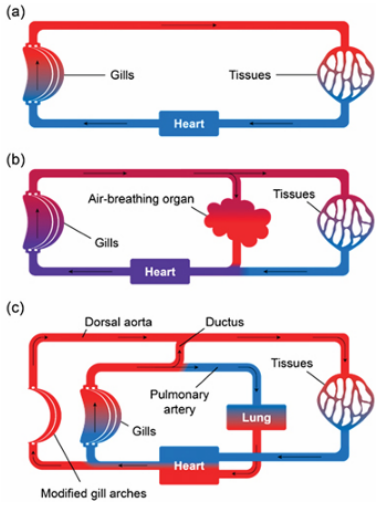


FIGURE 4.2 Generalized circulatory patterns found in (a) water-breathing fish, (b) air breathing fish, and (c) lungfish. The lungfish circulation shows aspects of the double circulation (i.e., respiratory and systemic circulations in parallel) found in mammals. Deoxygenated blood leaves the right side of the heart and travels to the gills and lung to be oxygenated, whereas oxygenated blood exits the left side and flows to the tissues via modified gill arches. Morphological modifications of the heart serve to separate oxygenated and deoxygenated blood.

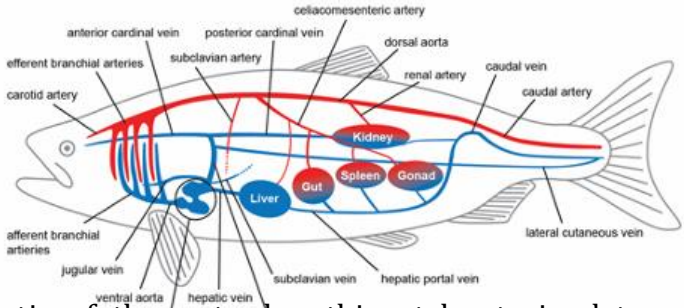


FIGURE 4.4 Schematic of the water-breathing teleost circulatory system. See fish vasculature for details. (Adapted from Sandblom, E. and Gräns, A., in Fish Physiology, Volume 36, Part A, The Cardiovascular System: Morphology, Control and Function, Academic Press, Cambridge, MA, 2017.)

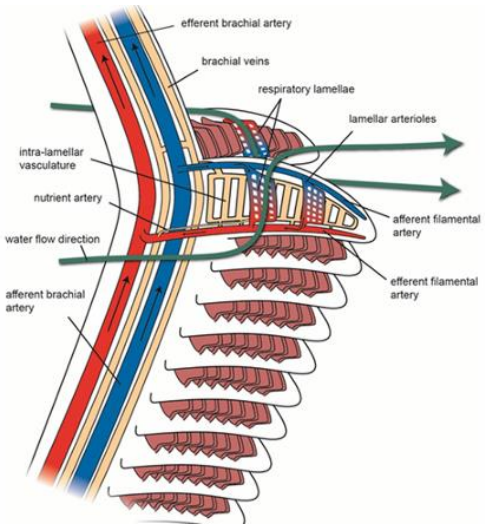


FIGURE 4.5 Schematic of blood flow through the gill. The countercurrent arrangement of the capillary bed and water flow (blood flows in the opposite direction to water) allows highly efficient oxygen uptake. (Adapted from Olson, K.R., J. Electron. Microsc. Tech., 19, 389–405, 1991.)

Fish have two portal systems: the hepatic portal system and the renal portal system (Figure 4.4). As in other vertebrates, the hepatic portal system drains blood from the stomach, intestines, spleen, and liver through the hepatic portal vein into the sinusoidal capillaries of the liver and back out through the hepatic veins to the sinus venosus. The renal portal system is unique to fishes; blood draining the trunk muscles, tail, and skin via the caudal vein, as well as the post-abdominal region (e.g. bladder and gonads), travels through the kidney before returning to the heart via the ducts of Cuvier.

Digestion

During digestion, oxygen consumption (termed specific dynamic action [SDA] or heat increment of feeding; see Chapter 6) increases substantially to support nutrient digestion, absorption, distribution, and processing. Concomitantly, gut blood flow (GBF) increases 40–150% from resting levels (30–40% of; Farrell et al. 2001) to support nutrient, hormone, and waste transport and to supply oxygen to the respiring intestinal tissues. The increased GBF is achieved via an increase in and/or a redistribution of regional blood flow to the stomach and intestinal tissues. Although several studies have focused on the mechanisms of GBF control and coordination, in general, regional blood flow and tissue prioritization remains an understudied area. However, the influence of exercise on GBF has been examined in a few species. In unfed fish, GBF decreases during swimming, presumably to prioritize oxygen delivery to the swimming muscles. However, in some species, GBF in fed fish is spared to maintain blood flow to the intestines (e.g. salmonids). In contrast, GBF decreases during swimming in fed sea bass. Several studies have examined how environmental factors (e.g. temperature, hypoxia, hypercapnia, salinity, and acute stress) influence GBF Given the importance of digestion for a fish's ability to grow and thrive, the influence of climate change on the integrated cardiovascular response during digestion is likely to be a fruitful area of future research.

High temperature

An extensive literature has considered how increasing temperature (across both acute and acclimation time scales) influences cardiovascular variables in fishes. Broadly speaking, increases with warming temperatures, which is almost exclusively driven by an increase in fH. In recent years, the heart has emerged as a potential mediator of thermal tolerance in fishes. Accordingly, one of the most active research areas in the field today seeks to determine why fish hearts fail at high temperature. One leading possibility is that the ion channels involved in E C coupling become functionally impaired at elevated temperatures, which would compromise the spread of the action potential across the heart and reduce Fh. One study with brown trout (Salmo trutta fario) found that Na+ channels were the most temperature sensitive. Another possibility

is that mitochondrial function could become impaired at high temperature, causing a limitation in ATP supply for the heart.

Low temperature

For freshwater fish that experience near-freezing temperatures in winter, cardiac physiology must be adjusted to accommodate the cold temperature driven effects on cardiac excitability and contractility, blood viscosity, and associated vascular resistance. The physiological strategy employed by fishes in response to cold temperature acclimation (or acclimatization) varies among species and is reflective of their overwintering strategy. As summarized in Table 4.1, some species, such as rainbow trout (Oncorhynchus mykiss) and burbot (Lota lota), exhibit physiological compensation that allows the continuation of an active lifestyle at cold temperature. By comparison, species that experience prolonged periods of oxygen deprivation during the winter months, such as the crucian carp (Carassius carassius), must prime physiological processes to conserve ATP, making positive compensatory changes maladaptive. Accordingly, for these species, cold exposure serves as an important cue to reduce activity, metabolic rate, and subsequently cardiac activity in anticipation of winter hypoxic and/or anoxic conditions. The phenomenon is termed inverse thermal acclimation. Despite these general distinctions, some cardiophysiological responses to cold are shared across overwintering strategies. For example, both rainbow trout and crucian carp exhibit increased density of K+ currents in the heart with cold acclimation, and the Alaska blackfish (Dallia pectoralis) displays a combination of down regulatory and cold-compensatory responses to cold acclimation. Finally, it is becoming increasingly recognized that naturally occurring seasonal acclimatization induces more pronounced changes in cardiac function than laboratory-based thermal acclimation.

Limiting oxygen levels

Hypoxia and anoxia are prevalent in a wide range of aquatic systems, both naturally and due to anthropogenic causes. For many water-breathing fish, the premier cardiovascular response to aquatic hypoxia is a cholinergically mediated reflex slowing of fH (termed hypoxic bradycardia). Hypoxic bradycardia is mediated through the activation of O2 chemoreceptors located in the gills; in most species, these chemoreceptors sense and respond to changes in oxygen in the inspired water. Although hypoxic bradycardia is hypothesized to have several direct benefits to the heart variation exists among species with regard to the water oxygen level at which hypoxic bradycardia is initiated as well as in the magnitude of fH depression. In fact, some species do not always exhibit hypoxic bradycardia (e.g., rainbow trout, winter flounder, Pseudopleuronectes americanus, and Atlantic cod, Gadus morhua whereas others do not exhibit hypoxic bradycardia at all (e.g., red- and white-blooded Antarctic fishes) even if exposed to anoxia (e.g., crucian carp). The variability relates to differences in ecology, lifestyle, and hypoxia tolerance, but intra-specific differences could also reflect differences in experimental design. The typical cardiac response of air-

breathing fish with biomodal respiration when exposed to aquatic hypoxia is a cycling of bradycardia prior to an air breath and a subsequent transient tachycardia (increase in heart rate) that commences with or just after each air breath (Graham 1997; Stecyk 2017). In addition to bradycardia, typical cardiovascular responses to hypoxia exposure include increased Rgill, PDA, and Rsys in water-breathing teleosts but unchanged or decreased PDA and Rsys in elasmobranchs (see Table 4.1 for definitions) (Stecyk 2017). In air-breathing fish, the prevailing response to aquatic hypoxia exposure with continued access to atmospheric air is an increase in, augmented blood flow to the air-breathing organ, and decreased blood flow to the gills. Nevertheless, despite these general trends, it is becoming increasingly clear that the cardiovascular responses to oxygen deprivation are highly variable among and even within species.

5. Iono- and Osmoregulation

General introduction

With the exception of some prokaryotes (e.g. halobacteria), metabolism in all cells has evolved to function optimally in a particular ionic milieu. Intracellular Na+ is low (~10 mM), K+ is high (~130 mM), and concentrations of divalent cations (Ca2+, Mg2+, etc.) are in the micro- to low millimolar range. Maintaining this ionic environment is critical because natural selection has optimized macromolecular structure and function under these conditions. Different strategies of achieving intracellular ion homeostasis have evolved to support cell function in environments with variable ion concentrations. Cells of fishes and other multicellular animals are isolated from the environment not only by their plasma membrane but also by an epithelial cell layer that separates the body fluids from the external milieu (Figure 5.1). This additional layer buffers environmental salinity fluctuations at the level of body fluids, shifting some of the energetic burden of maintaining intracellular ion homeostasis from individual cells to the level of specialized epithelial tissues that are in direct contact with the external milieu. Most fishes, except primitive taxa, pursue this strategy of osmoregulation to maintain intracellular ion homeostasis.

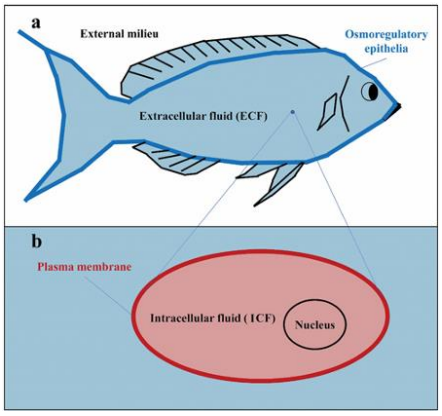


FIGURE 5.1 The osmotic and ionic milieu is controlled at two barriers. (a) The outer barrier (blue line) reflects epithelia that, in most fishes, maintain an osmotic gradient

between the external environment and the extracellular fluid (ECF). (b) The inner barrier (red line) is the plasma membrane surrounding each cell, which maintains strong ionic (but not osmotic) gradients (e.g. [Na+] and [K+]) between ECF and intracellular fluid (ICF).

Evolutionary strategies

Fishes encompass a spectrum of strategies ranging from osmoconformers to osmoregulators (Table 5.1; Figure 5.2). In osmoconformers, ionic homeostasis is regulated primarily at the level of cells. This strategy is evolutionarily primitive and employed by hagfish. However, lamprey (the other clade of jawless fishes) are osmoregulators, a more advanced strategy, as are teleosts (bony fish). Elasmobranchs use a third approach, being osmoconformers but strong ionoregulators. These strategies are briefly discussed in the following subsections.

HAGFISH

The body fluids of these primitive marine fish are approximately isosmotic with seawater (SW), whereas their ionic composition is similar but not identical to that of SW. Hagfish represent an early evolutionary stage at which the burden of maintaining intracellular ion homeostasis has started to shift from the plasma membrane of cells to the level of specialized epithelia. These epithelia provide a barrier between the fish and the environment, allowing the ionic composition of the body fluids to be controlled. However, ion differences across hagfish external epithelia are small because ionoregulatory mechanisms are at a primitive evolutionary stage. Specialized epithelial cells (ionocytes) probably evolved to support acid–base homeostasis and/or nitrogenous waste excretion rather than ionoregulation. Hagfish have ionocytes in their osmoregulatory epithelia and may represent an evolutionary transition in which ionocytes start to be adopted for ionoregulation beyond their other roles.

LAMPREY

Unlike hagfish, lamprey have conquered freshwater (FW), with some species remaining in FW throughout their life cycle, while others develop in FW but spend most of their life in SW. The evolutionary transition from SW to FW requires effective ionoregulation because FW ion concentrations are too low to be tolerated in the body fluids. Thus, FW invasion required the evolution of mechanisms to maintain significant ionic and osmotic gradients across external epithelia. As in teleosts and tetrapods, the body fluid osmolality of FW lamprey is 300–350 mosmol/kg. To achieve hyperosmotic body fluids relative to FW, lamprey actively accumulate NaCl and excrete water, processes that are energy dependent. Active NaCl uptake counteracts passive diffusional NaCl loss from the body fluids to FW and is achieved using ionocytes in the branchial epithelium. Thus, one evolutionary advance that appeared in the ancestor of extant lamprey was the functional adoption of branchial ionocytes for iono- and osmoregulation. A second, equally important advance was the adoption of renal

glomeruli and distal tubules for selective filtration and excretion of water. These renal novelties were critical for maximizing NaCl reabsorption and retention.

ELASMOBRANCHS

Elasmobranchs include sharks, skates, and rays, and most species are marine, although several euryhaline species complete at least part of their life cycle in FW. Marine elasmobranchs are strong ionoregulators while still being osmoconformers. They maintain an NaCl concentration of about one third SW yet a total osmolality of ~1000 mosmol/kg, slightly hyperosmotic to SW, in their body fluid. Organic osmolytes, primarily urea and trimethylamine oxide (TMAO), bridge this 'osmotic gap'. The passive inward diffusion of NaCl is countered by active NaCl excretion by ionocytes in the rectal gland. Marine elasmobranchs must also actively retain elevated urea and TMAO. To counter diffusional loss of these organic osmolytes to SW, they are constantly produced as by-products of nutrient metabolism.

Euryhaline elasmobranchs are iono- and osmoregulators in FW, with high body fluid osmolality (>600 mosmol/kg) due to urea and TMAO concentrations that are reduced in FW but remain considerable. The high osmolality of elasmobranch body fluids relative to FW presents a significant energetic burden; that is, in FW, elasmobranchs would have to either greatly tighten their epithelia to minimize diffusion and/or actively absorb NaCl against huge concentration and osmotic gradients. Unlike for teleosts, it is currently unknown whether euryhaline elasmobranchs regulate the leakiness of epithelial tight junctions when entering FW. Moreover, the number of euryhaline elasmobranchs able to enter FW for extended periods is small relative to their strictly marine counterparts. It appears that the energetic burden associated with maintaining elevated body fluid osmolality prevents widespread evolutionary expansion of elasmobranchs into FW habitats. The few stenohaline FW elasmobranchs (e.g. the Amazonian stingray Potamotrygon sp.) have low urea levels similar to those of FW teleosts.

TELEOSTS

All teleosts are osmoregulators, with body fluid osmolality of ~300–350 mosmol/kg in both FW and SW. In an evolutionary sense, this strategy shifted much of the burden of maintaining intracellular ion homeostasis from the level of the cell to that of specialized epithelia. For instance, instead of a Na+ gradient across the plasma membrane of ~500 mosmol/kg in SW versus 10 mosmol/kg intracellular, it becomes ~150 mosmol/kg in body fluids versus 10 mosmol/kg intracellular. However, maintaining body fluid osmolality below that of SW represents a formidable task for marine teleosts. Osmotic water loss across the external epithelia is countered by ingestion of SW. Water absorption across the intestine is driven by NaCl absorption, which adds to the NaCl load gained by diffusion. Excess NaCl is actively excreted by gill ionocytes, while the absorbed water is actively retained by reabsorption in the renal tubule, keeping urinary volume to the minimum necessary for waste and divalent ion

excretion. This process yields a net gain of water that compensates for the dehydration resulting from osmosis. Drinking rates of euryhaline fishes vary with environmental salinity, being virtually zero in FW and high in SW. Maintaining body fluid osmolality above that of FW similarly represents a challenge for FW teleosts. Water gain by osmosis across the external epithelia is countered by copious production of dilute urine, which is not completely devoid of ions and aggravates ion loss. Thus, gill ionocytes in FW teleosts actively take up NaCl and Ca2+

Physiology of iono- and osmoregulatory tissues

The main iono- and osmoregulatory tissues are skin (especially during early development), gill, renal, and gastrointestinal epithelia. Other epithelia (e.g. those of the operculum, oral cavity, and pyloric caeca) may also have osmoregulatory functions but largely represent structural and functional extensions of gill or gastrointestinal epithelia and are not considered separately in the following overview. The elasmobranch rectal gland is an extension of the gastrointestinal tract (GIT) but functions like the gill of a marine teleost and will be discussed briefly in that context.

Skin

In most adult fishes, ion transport across the body surface is dominated by the gill, and the skin serves as a permeability barrier. The exceptions include specific skin regions that are ionocyte rich, such as the opercular epithelium, and a few marine and amphibious species that have ionocytes across the body surface. The opercular epithelium was used as a model to investigate NaCl secretion in marine teleosts because, unlike the gill, it provided a flat epithelium that could be used for Ussing chamber analysis of electrophysiology, and ionocytes are readily accessible to electrodes.

In embryonic and larval fishes, ion transport is strictly cutaneous until the gill develops. Systemic regulation of ion and water balance begins during the early embryonic stages, when ionocytes appear in the skin and yolk sac epithelium. These cutaneous ionocytes increase in abundance as development proceeds, peaking as ionocytes first appear in the developing gill. As the gill develops, the abundance of branchial ionocytes increases, while that of cutaneous ionocytes falls (Figure 5.3).

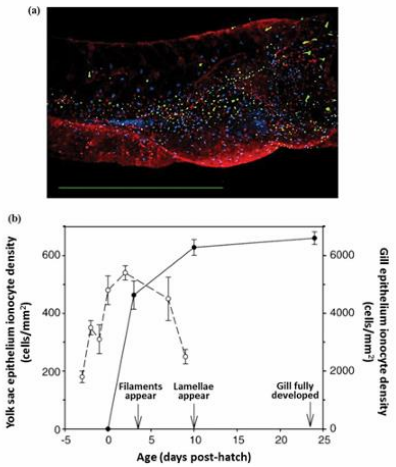


FIGURE 5.3 (a) Light micrograph of the lateral body and yolk sac of a larval zebrafish (Danio rerio) at 4 days-post-fertilization. Immunofluorescent markers identify three ionocyte types: green cells take up Na+ and excrete H+, red cells take up Na+ and Cl−, and blue cells take up Ca2+. Scale bar = 1 mm. (b) Changes in cutaneous versus branchial ionocyte abundance during early development of tilapia (Oreochromis mossambicus). (Panel a: image courtesy of E. Kunert. Panel b: figure modified from Rombough, P., Comp. Biochem. Physiol. A, 148, 732–742, 2007.)

Gills

The fish gill is a structurally complex, multifunctional organ that serves as the primary site of gas transfer as well as body fluid ion, water and pH regulation and nitrogenous waste excretion. Anatomically, the gills consist of paired arches on the pharynx that support rows of filaments, from which arise small, plate-like lamellae. Blood flowing through the lamellae is separated from water flowing over them by a branchial epithelium that provides the extensive surface area, high permeability and thin diffusion barrier necessary for efficient gas transfer. However, the structural features that enhance gas transfer also favour diffusive ion and water movements. Gill morphology therefore represents an 'osmorespiratory compromise' in which lamellar surface area and the thickness of the lamellar epithelium have been selected through evolutionary time to balance the minimum rate of O2 transfer needed for metabolism against the maximum rate of passive ion and water transfer that can be tolerated. To manage the osmorespiratory compromise, some fish species use reversible gill remodelling. With this strategy, the lamellar epithelium is covered by a cell mass to reduce passive ion and water movement when O2 availability is high and O2 demand is low, and the cell mass is shed to increase O2 uptake in response to exercise, hypoxia or warm temperatures.

Freshwater Fishes

Classic studies of ion uptake in FW fishes (reviewed by Evans et al. 2005; Evans 2011; Gilmour and Perry 2009) indicated that Na+ uptake is linked to H+ secretion and Cl− uptake is linked to HCO3 − secretion. As CO2 crosses the gill by diffusion, some portion is hydrated to H+ and HCO3 −, which serve as counter-ions in ion uptake mechanisms. Due to this coupling of NaCl uptake to secretion of an acid–base equivalent, the FW gill is essential for acid–base regulation as well as ion balance. Ionocyte contributions to ion uptake have been recognized since the early 1960s, but the last 15 years have yielded major advances. Through immunohistochemistry and in situ hybridization, transporters have been localized to specific cell populations. Techniques such as scanning ion-selective electrodes and isolation of specific cell populations by magnetic separation or laser capture microdissection provided insight into ionocyte function. The coupling of such approaches with loss-of function technologies proved particularly powerful (Horng et al. 2007). Across the handful of species examined to date, there are common features but also marked differences. What

is clear, however, is that multiple ionocyte types differing in transporter complement and hence function is present in the branchial epithelium (Figure 5.4).

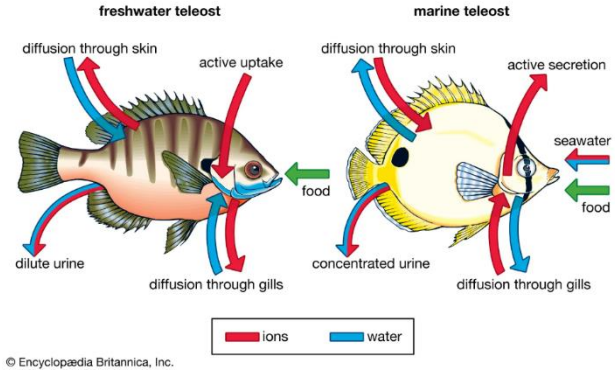


FIGURE 5.4 Generalized osmotic regulation in freshwater and marine teleost fishes

Marine Fishes

The branchial epithelium of marine teleosts contains an NaCl-secreting ionocyte. Unlike FW ionocytes, which generally occur singly in the branchial epithelium and form extensive, multi-stranded tight junctions with neighboring cells, SW ionocytes occur in multicellular complexes with accompanying accessory cells. The apical membranes of these cells form a recessed crypt, which is a distinctive feature of the marine teleost branchial epithelium. Shallow tight junctions occur between ionocytes and accessory cells within the complex and are selectively leaky to cations, a structural feature that is of functional importance for Na+ secretion.

Salt secretion is achieved by secondary active transport of Cl− and passive diffusion of Na+. The mechanism (Figure 5.4) relies on basolateral NKA, which keeps intracellular Na+ levels low to create a gradient for Cl− entry coupled to Na+ via basolateral Na+, K+, 2Cl− cotransporter (NKCC). Chloride ions exit the ionocyte apically via a cystic fibrosis transmembrane conductance regulator (CFTR) Cl− channel, while Na+ leaves paracellularly via the cation-selective leaky tight junction. The final component of the mechanism is the recycling of K+ ions that entered the cell via NKA and NKCC by a K+ channel.

Kidney

In fishes, the gill is responsible for the majority of active transepithelial ion transport, but the kidney contributes to osmoregulation as well as waste excretion, detoxification, hormone production and acid–base regulation. The two main osmoregulatory functions of fish kidneys are water excretion (urine production) and the secretion of divalent ions (chiefly Mg^{2+}, Ca^{2+}, PO_4^{3-} and SO_4^{2-}).

Euryhalinity

It is evident that FW and marine fishes face different osmoregulatory challenges. Energy-dependent water and salt transport rely on disparate sets of proteins, molecular transport mechanisms and cellular adaptations in FW versus marine fishes. Ionocytes

differentiate into distinct populations depending on the external salinity. Furthermore, intercellular junctions, epithelial morphology, blood circulation, drinking rates and endocrine functions are markedly different in FW and marine fishes. Stenohaline fishes are limited in their ability to overcome these differences; they are adapted to either FW or SW and cannot tolerate salinity deviations that exceed 15 g/kg from the norm (Figure 5.7). In contrast, euryhaline fishes tolerate both FW and SW. These species are fewer in number than stenohaline species and have multiple independent evolutionary origins. Even within families and genera, a mosaic phylogenetic pattern of euryhalinity is evident.

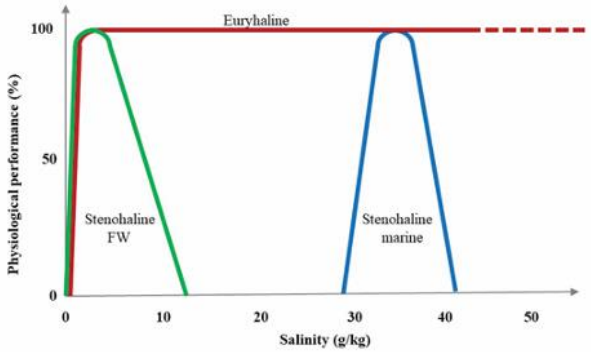


FIGURE 5.7 Tolerance limits of stenohaline marine (blue line), stenohaline freshwater (FW, green line) and euryhaline fishes (red line). Euryhaline species typically exceed the upper salinity tolerance limit of marine fishes, tolerating up to 2–4× sea water.

Activity

1. Illustrate and Identify the External and Internal fish morphology

Post-assessment

1. What are the two (2) major classifications of fish locomotion? (5pts)
2. Differentiate stenohaline and euryhaline (5pts)
3. What are the parts of the gills of a fish? (5pts)
4. Explain how marine and freshwater fish osmoregulation works? (5pts)

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1.4 Acknowledgement

I would like to express my sincere gratitude to all those who contributed to the creation and development of these instructional materials. Their guidance, expertise, and support have been invaluable in the completion of this learning packet. I am deeply appreciative of the resources, insights, and feedback that have shaped these materials

and made them more comprehensive and accessible. Special thanks to the Samar State University Mercedes Campus Director Prof. Jerson C. Sorio and to our Head for Instructions Prof. Noeme D. Pajarillo for their continuous encouragement and assistance throughout this process.

Especially also to our almighty god for the guidance, strength, and inspiration upon completing this instructional material.

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Note:

For your outputs, questions, and clarifications, kindly send it to



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